RAILWAY RESERVATION SYSTEM

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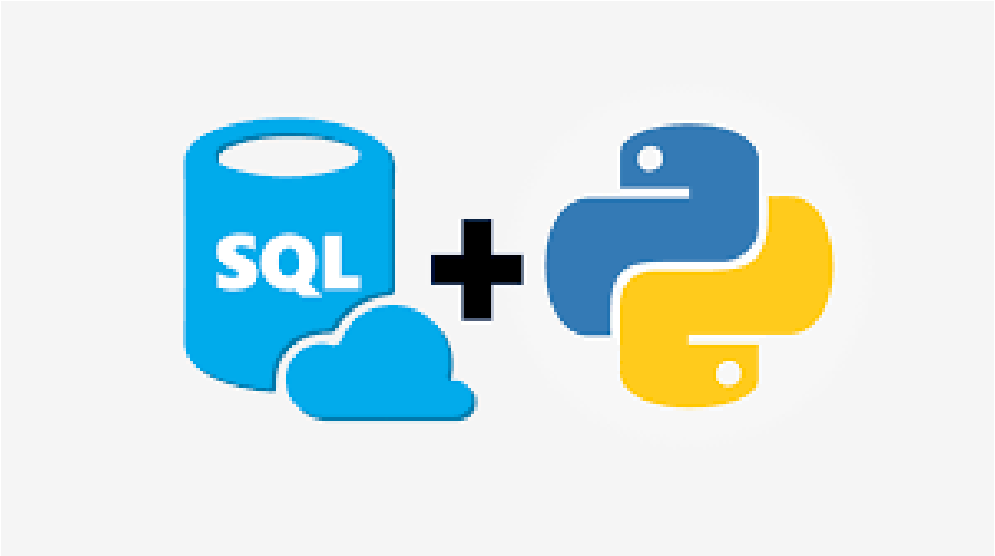
HARDWARES AND SOFTWARE REQUIRED

HARDWARE

1.Desktop/Laptop

SOFTWARE

1.Python (latest version)
2.MySQL
3.Python Connector Module



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INTRODUCTION

The project in on RAILWAY RESERVATION SYSTEM

Objectives of the Project:

1.Administrative Empowerment

The project aims to cater to the intricate requirements of both administrators and passengers within the railway ecosystem. The core objectives include:

- **User Registration & Authentication**:
  The system implements a streamlined user registration process, requiring users to provide essential details such as user ID, name, phone number, email ID, and password. All data is securely stored in a MySQL database. Robust authentication mechanisms ensure the security and integrity of user accounts.
- **Administrative Functions**:
  The Admin Panel serves as the backbone of the system, offering vital tools such as:
  - **Add Train**: Enables administrators to add new trains with details like train number, name, origin, destination, journey distance, time, seat availability, and fares.
  - **Update Train Details**: Facilitates modifications to existing train details, including seat availability, journey schedules, and fare adjustments, to meet dynamic demands.
  - **Cancel Train**: Allows administrators to cancel trains in response to operational or demand changes, ensuring system flexibility and adaptability.

2. Enhancing Customer Experience

The Customer Panel is designed with user convenience as a priority, providing an intuitive interface for seamless ticket booking and management.

- **User Interaction**:
  - **User Login**: A secure login interface requiring user ID and password ensures account safety.
  - **Forgot User ID**: Users can retrieve their forgotten user IDs via their registered email, enhancing accessibility.
- **Passenger Panel Functionalities**:
  - **Train Search**: Users can search for trains by origin and destination, accessing details such as schedules, seat availability, and fares.
  - **Book Tickets**: Facilitates ticket reservations by allowing users to select trains, classes, and passengers.

Cancel Tickets: Provides an easy process for users to cancel bookings, offering flexibility in managing travel plans.

Key Code Functionalities

1. Integrated User Management Functions

- **New User Registration**: The `new_user()` function systematically registers new users, generates unique user IDs, and stores data in the MySQL database.
- **Forgot User ID**: The `forgot_user_id()` function retrieves registered user IDs via email verification, ensuring a smooth user experience.
- **Old User Authentication**: The `old_user()` function verifies user credentials, granting access to system functionalities for registered users.

2. Admin Panel Operations

- **Admin Password Verification**: The `adminpassword()` function ensures secure access to the admin panel through a password verification system.
- **Train Management Operations**: Functions like `add_train()`, `update_details()`, and `cancel_train()` empower administrators to manage train details effectively, including adding new trains, updating information, and canceling services.

3. Passenger Panel Operations

- **Train Search**: The `Train_Search()` function enables users to search for trains by origin and destination, displaying comprehensive details.
- **Book Tickets**: The `Book_Ticket(uid)` function facilitates ticket reservations, allowing users to select trains, classes, and passengers.

- **Cancel Tickets**: The `Cancel_Ticket()` function simplifies the process of canceling reservations, ensuring flexibility for users.

Significance of the Project

This project holds significant real-world value by addressing the critical need for efficient and adaptive transportation systems. An effective railway reservation system is essential in today's fast-paced world, where efficient transportation is pivotal. By prioritizing user satisfaction, security, and transparency, the project serves as a robust solution for the evolving demands of the railway industry.

PYTHON SOURCE CODE :

```
import random
from tabulate import tabulate
import mysql.connector as con

# Establish global connection details
def get_db():
    return con.connect(
        host="localhost",
        user="root",
        password="admin",
        database="train_reservation"
    )

# --- NEW USER REGISTRATION SECTION ---
def new_user():
    dbo = get_db()
    co = dbo.cursor()
    pid = random.randint(0, 1000) * 10
    print("---------------------------------------------------------------- ")
    print(" \n Welcome to our reservation system \n Register Yourself here to use our system")
    uid = input("Enter your user id: ")
    name = input("Enter your name: ")
    pno = input("Enter your phone no: ")
    eid = input("Enter your email_id: ")
    pwd = input("Enter your password: ")

    query = "insert into user values ('{}',{},'{}',{},'{}','{}')".format(uid, pid, name, pno, eid, pwd)
    co.execute(query)
    dbo.commit()
    print("************* Congratulations!!! Your id is successfully created **************")
    print("---------------------------------------------------------------- ")
    dbo.close()

# --- FORGET USER ID ---
def forgot_user_id():
    dbo = get_db()
    co = dbo.cursor()
    email = input("Enter your registered email: ")
    co.execute("select user_id from user where email_id like '{}'".format(email))
    emel = co.fetchall()
    if emel:
        for i in emel:
            print("Your user_id is: ", (i[0]))
    else:
        print("Email not found.")
    dbo.close()

# --- OLD USER LOGIN ---
def old_user():
    dbo = get_db()
    co = dbo.cursor()
    print("\n---------------------------------------------------------- \n")
    uid = input("Enter your user id: ")
    co.execute("select user_id from user where user_id = '{}'".format(uid))
    b = co.fetchall()

    if len(b) == 0:
        print("---------------- Your given id is not registered ----------------")
        print("1. Try again\n2. Forgot user id\n3. Register as a new user")
        choose = int(input("Choose an option: "))
        if choose == 1: old_user()
        elif choose == 2: forgot_user_id()
        elif choose == 3: new_user()
```

```
    else:
        pas = input("Enter your password: ")
        co.execute("select password from user where user_id = '{}'".format(uid))
        n = co.fetchone()
        if n and pas == n[0]:
            print("\n---------------- Welcome back! ----------------\n")
            passenger_panel(uid)
        else:
            print("Incorrect Password!")
    dbo.close()

# --- ADMIN FUNCTIONS ---
def adminpassword():
    password = input("Enter admin password: ")
    if password == "CLASS12CS":
        print("****************** Access Granted ********************")
        admin_panel()
    else:
        print("*************** ACCESS DENIED ************")

def add_train():
    dbo = get_db()
    co = dbo.cursor()
    print("---------------------------------------------------------------- ")
    a = int(input("Enter train no: "))
    b = input("Enter train name: ")
    c = input("Enter train origin: ")
    d = input("Enter train destination: ")
    e = int(input("Enter train journey distance: "))
    g = input("Enter train journey time: ")
    h = int(input("Enter no of seats in AC: "))
    i = int(input("Enter no of seats in SL: "))
    j = int(input("Enter no of seats in GEN: "))
    k = int(input("Enter price of AC: "))
    l = int(input("Enter price of SL: "))
    m = int(input("Enter price of GEN: "))
    n = input("Enter days available: ")

    query = "insert into train_schedule values ({},'{}','{}','{}',{},'{}',{},{},{},{},{},{},'{}')"
    co.execute(query.format(a, b, c, d, e, g, h, i, j, k, l, m, n))
    dbo.commit()
    print("********* Train added successfully ************")
    dbo.close()

def update_details():
    dbo = get_db()
    co = dbo.cursor()
    while True:
        print("\n****** Update Train System ******")
        print("1. Train No\n2. Name\n3. Origin\n4. Destination\n5. Dist\n6. Days\n7. Time\n8. AC Seats\n14.
Exit")
        x = int(input("Choice: "))
        if x == 14: break

        tno_search = int(input("Enter existing Train No to update: "))

        if x == 1:
            new_val = int(input("New Train No: "))
            co.execute("update train_schedule set train_no=%s where train_no=%s", (new_val, tno_search))
        elif x == 2:
            new_val = input("New Train Name: ")
            co.execute("update train_schedule set train_name=%s where train_no=%s", (new_val, tno_search))
        # ... Add other elifs here following the same pattern ...
```

```
        dbo.commit()
        print("Updated successfully!")
    dbo.close()

def cancel_train():
    dbo = get_db()
    co = dbo.cursor()
    ct = int(input("Enter train no to delete: "))
    co.execute("delete from train_schedule where train_no={}".format(ct))
    dbo.commit()
    print("********** Train Deleted ****************")
    dbo.close()

def admin_panel():
    while True:
        print("\n****** Welcome to Admin Panel ******")
        print("1. Add Train\n2. Update Details\n3. Cancel Train\n4. Log out")
        try:
            opt = int(input("Choose option: "))
            if opt == 1: add_train()
            elif opt == 2: update_details()
            elif opt == 3: cancel_train()
            elif opt == 4: break
        except:
            print("Invalid Input.")

# --- PASSENGER FUNCTIONS ---
def Train_Search():
    dbo = get_db()
    co = dbo.cursor()
    o = input("Enter origin: ")
    d = input("Enter destination: ")
    co.execute("select * from train_schedule where origin like %s and destination like %s", (f"%{o}%",
f"%{d}%"))
    rows = co.fetchall()
    if rows:
        for row in rows:
            print(f"\nTrain: {row[1]} ({row[0]}) | Route: {row[2]} to {row[3]}")
            print(f"AC Fare: {row[9]} | SL Fare: {row[10]} | GEN Fare: {row[11]}")
    else:
        print("No trains found.")
    dbo.close()

def Book_Ticket(uid):
    dbo = get_db()
    co = dbo.cursor()
    co.execute("select train_no, train_name, origin, destination from train_schedule")
    r = co.fetchall()
    print(tabulate(r, headers=['No', 'Name', 'From', 'To'], tablefmt="fancy_grid"))

    trainno = int(input("\nEnter Train no: "))
    tcktno = int(input("Number of seats: "))
    print("1. AC | 2. SL | 3. GEN")
    typ = int(input("Class: "))

    col = "ac_fare" if typ == 1 else "sl_fare" if typ == 2 else "gen_fare"
    co.execute(f"select {col} from train_schedule where train_no={trainno}")
    price = co.fetchone()[0]

    for i in range(tcktno):
        cus = input(f"Passenger {i+1} Name: ")
        age = int(input("Age: "))
```

```
        gender = input("Gender (M/F): ")
        pnr = random.randint(100000, 999999)
        co.execute("insert into booked_tickets values (%s,%s,%s,%s,%s,%s,%s,%s)",
                (uid, pnr, trainno, cus, age, gender, price, "Confirmed"))

    dbo.commit()
    print(f"Total Amount: {tcktno * price}. Booking Successful!")
    dbo.close()

def Cancel_Ticket():
    dbo = get_db()
    co = dbo.cursor()
    pnr = int(input("Enter PNR to cancel: "))
    co.execute("delete from booked_tickets where pnr_no=%s", (pnr,))
    dbo.commit()
    print("Ticket Cancelled.")
    dbo.close()

def passenger_panel(uid):
    while True:
        print("\n****** Passenger Panel ******")
        print("1. Search Train\n2. Book Ticket\n3. Cancel Ticket\n4. Log out")
        choic = int(input("Choice: "))
        if choic == 1: Train_Search()
        elif choic == 2: Book_Ticket(uid)
        elif choic == 3: Cancel_Ticket()
        elif choic == 4: break

def user_panel():
    print("1. Register\n2. Login\n3. Back")
    out = int(input("Choice: "))
    if out == 1: new_user()
    elif out == 2: old_user()

# --- MAIN MENU ---
def main_menu():
    while True:
        print("\n*************** WELCOME TO TRAIN SYSTEM ***************")
        print("1. Admin Panel\n2. Passenger Panel\n3. Exit")
        try:
            ch = int(input("Choice: "))
            if ch == 1: adminpassword()
            elif ch == 2: user_panel()
            elif ch == 3: break
        except ValueError:
            print("Please enter a number.")

if __name__ == "__main__":
    main_menu()
```

MySQl DATABASE:

TRAIN TABLE:

```
mysql> select * from train_schedule;
```

train_no	train_name	origin	destination	journey_distance	total_time	ac1	sl	gen	ac1_fare	sl_fare	gen_fare	avl_days
11020	KONARK EXPRESS	Bhubaneswar	CSMT	1930	36HR 40MIN	120	250	100	2400	950	200	S M T W TH F S
11209	HIRAKUD EXP	BALASORE	BHADRAK	200	5HR 55MIN	85	65	90	1900	500	200	S M T W TH F S
12321	MUMBAI MAIL	HOWRAH JN	CSMT	2159	37HR 40MIN	128	150	50	2850	850	350	S M T W TH F S
12457	HOWRAH EXP	CSMT	DELHI	4500	55HR 54MIN	58	56	54	2600	300	365	S M TH W S
12472	SWARAJ EXPRESS	JAMMU	BANDRA TERMINUS	2028	30HR 18MIN	256	182	90	2650	650	450	S M T W TH F S
12809	HOWRAH EXPRESS	CSMT	HOWRAH	1968	32HR 55MIN	90	350	80	2300	800	300	S M T W TH F S
12841	COROMONDAL EXPRESS	SHALIMAR	CHENNAI	1659	25HR 35MIN	125	100	90	2260	760	360	S M T W TH F S
20807	HIRAKUD EXPRESS	VISAKHAPATNAM	AMRITSAR	2570	38HR 45MIN	156	220	68	3400	700	100	S M T W TH F S
20861	PURI ADI SF	PURI	SURAT	1977	31HR 05MIN	128	240	200	2900	900	250	S M T W TH F S
123432	PRASANTI EXPRESS	Puri	Chennai	1980	40HR 30 MIN	123	234	345	3000	2000	1000	S M T W TH F S

```
10 rows in set (0.02 sec)
```

USER DETAILS :

user_id	pass_id	pass_name	ph_no	email_id	password
asish	7830	Asish Kumar	814427568	asish@gmail.com	asish@123
john@doe	6990	John Doe	5551234	john.doe@gmail.com	john@doe
kalia	8810	chintu	2514584	roshni133@gmail.com	789
LP	8230	lokashree	789956784	lokashree18@gmail.com	panipuri
mohit	8910	n.mohit	36533	mohit@123	mohit123
om	6600	Omm panda	78942665	omm@gmail.com	omm@panda
Sheetal	2420	Sheetal sahu	456321456	sheetal19@gmail.com	sheetal1906
swati	7510	ssp	451024526	swati09@gmail.com	topper
vinit	3000	avk18	68546894	avk18@gmail.com	chal

BOOKED_TICKETS:

user_id	pnr_no	train_no	customer_name	age	gender	price	ticket_status
asish	794470	11020	hgdfhgt	46	m	200	Confirmed
vinit	926193	12321	LIZA	22	F	850	Confirmed
vinit	312519	12321	PRAMILA	48	F	850	Confirmed
vinit	203328	12321	MIRA	45	F	850	Confirmed
swati	964758	20861	swatismita pradhan	17	f	900	Confirmed
vinit	662927	11020	sameer	17	m	2400	Confirmed
vinit	161090	11020	basundhara	16	f	2400	Confirmed
Sheetal	141050	12472	Sheetal Sahu	18	F	2650	Confirmed

OUTPUTS:

TRAIN RESERVATION MAIN PAGE:

```
-----------------------------------------------------
********WELCOME TO TRAIN RESERVATION SYSTEM********
1. Admin panel
2. Passenger panel
3. Exit
Enter your choice: 1
Enter your password: CLASS12CS
******************Access Granted********************
-----------------------------------------------------------------
-----------------------------------------------------------------
******Welcome to admin panel******
1. Add train
2. Update details
3. Cancel Train
4. Log out
Choose your option: 1
-----------------------------------------------------------------
Enter train no: 12347
Enter train name: RAJDHANI EXP
Enter train origin: DELHI
Enter train destination: JAMMU KASHMIR
Enter train journey distance: 1200
Enter train journey time: 9 HR 30MIN
Enter no of seats in AC: 290
Enter no of seats in SL: 200
Enter no of seats in GEN: 300
Enter price of AC: 3900
Enter price of SL: 900
Enter price of GEN: 500
Enter days available: S M T W TH F S
-----------------------------------------------------------------
*********You have added a new train details successfully************
-----------------------------------------------------------------
```

THE ADMIN LOGIN PANEL:

```
******Welcome to admin panel******
1. Add train
2. Update details
3. Cancel Train
4. Log out
Choose your option: 2
-----------------------------------------------------------------------
THIS IS OUR ALL TRAINS AVAILABLE
-----------------------------------------------------------------------
```

train_no	train_name	origin	destination
11020	KONARK EXPRESS	Bhubaneswar	CSMT
11209	HIRAKUD EXP	BALASORE	BHADRAK
12321	MUMBAI MAIL	HOWRAH JN	CSMT
12347	RAJDHANI EXP	DELHI	JAMMU KASHMIR
12457	HOWRAH EXP	CSMT	DELHI
12472	SWARAJ EXPRESS	JAMMU	BANDRA TERMINUS
12809	HOWRAH EXPRESS	CSMT	HOWRAH
12841	COROMONDAL EXPRESS	SHALIMAR	CHENNAI
20807	HIRAKUD EXPRESS	VISAKHAPATNAM	AMRITSAR
20861	PURI ADI SF	PURI	SURAT
123432	PRASANTI EXPRESS	Puri	Chennai

```
-----------------------------------------------------------------------
```

CANCEL TICKET:

```
-----------------------------------------------------------------
******Welcome to passenger panel******
1. Train Search
2. Book Ticket
3. Cancel Ticket
4. Log out
Enter your choice to use: 3
enter pnr no. you want to cancel: 545868
***********  Your ticket cancelled Successfully ****************
-----------------------------------------------------------------
```

BOOKING TICKETS :

```
Enter your Train no: 12321
Enter no of seats you want to book: 2
1. AC TICKET
2. SL TICKET
3. GEN TICKET
Enter your choice of class: 1
2850
Enter customer name: snehashish
Enter your age: 18
---------------- For gender M=Male, F=Female, O=Other ----------------
Enter your gender: m
Your PNR no is:  216440
Enter customer name: bansika
Enter your age: 17
---------------- For gender M=Male, F=Female, O=Other ----------------
Enter your gender: f
Your PNR no is:  545868
Your total ticket price is:  5700
-----------------------------------------------------------------------
```

SEARCH TRAIN:

```
Enter your origin: csmt
Enter your destination: delhi
Train no.:  12457
Train name:  HOWRAH EXP
Origin:  CSMT
Destination:  DELHI
Journey distance:  4500
Available days:  S M TH W S
total time:  55HR 54MIN
Seats in ac1:  58
Seats in sl:  56
Seats in GEN:  54
Fare of ac1:  2600
Fare of sl:  300
Fare of gen:  365
```

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